THARUN G

Geospatial Data Engineer | Remote Sensing & GIS

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SUMMARY

Third-year CSE student combining machine learning with geospatial data: Landsat and Sentinel analysis in Google Earth Engine, spatial ML with GeoPandas and PySAL, and web GIS platforms. Completed a spatial data engineering internship at Thazhal Geospatial. Seeking GIS, remote sensing or geospatial data roles.

EDUCATION

B.E. Computer Science and Engineering | National Engineering College, Kovilpatti 2024 – 2028 (Expected)

CGPA: 6.8 / 10

SKILLS

Remote Sensing & GIS: Google Earth Engine, QGIS, ArcGIS, GDAL, Spatial Analysis, Earth Observation, Satellite Imaging (Landsat, Sentinel-1/2), NDVI/NDBI

Spatial ML: GeoPandas, PySAL, Random Forest, scikit-learn, Pandas, NumPy

Development: Python, TypeScript, Next.js, SQL, Cloudflare D1/R2, AWS, Git

EXPERIENCE

Spatial Data Engineering Intern | Thazhal Geospatial June 2026

- Built a full-stack spatial AI tool (Next.js, Python, GeoPandas) that estimates Total Addressable Market for financial-infrastructure expansion.
- Wrote offline pipelines to ingest, clean and spatially aggregate 23,000+ commercial locations from the Google Places API into municipal boundaries.
- Applied PySAL spatial lag and Random Forest models to account for competitor density when scoring candidate locations.

PROJECTS

GeoForesight: Remote Sensing & GIS Research Platform | Next.js, TypeScript, Tailwind CSS, Cloudflare D1/R2

- Built a web platform where remote-sensing research teams publish, annotate and peer-review geospatial findings in one place.
- Implemented role-based moderation, threaded comments and Cloudflare D1/R2 storage.

Dubai Multitemporal Urban Analysis (academic lab) | Google Earth Engine, Landsat 8/9

- Computed NDVI, NDBI and RGB composites from Landsat 8/9 imagery for Dubai (2015–2024) to study vegetation and built-up change.

GeoAI Urban Expansion & Heat Risk Decision Support System (capstone) | Google Earth Engine

- Designing a decision-support system for Dubai that combines urban-expansion and heat-risk indicators from satellite data.

Conflict-Zone Damage Assessment | Google Earth Engine, Sentinel-1/2

- Used Sentinel-1 and Sentinel-2 imagery in Google Earth Engine to assess damage across conflict zones.

Astrovia: Explainable Satellite Anomaly Detection | Python, PyTorch, scikit-learn, SatNOGS, Streamlit, Plotly, Docker SIH 2026 (Waitlisted)

- Replaced fixed-threshold telemetry alarms with per-channel PyTorch LSTM forecasters and z-score scoring of prediction residuals; validated on NASA SMAP and MSL (Curiosity) datasets.
- Built an explainability layer based on prediction-error attribution that ranks root-cause sensors without the compute cost of SHAP.

Smart Urban Mobility & Traffic Intelligence Dashboard | Bengaluru

- Built a business-intelligence dashboard with transit, traffic, weather and demographics modules for urban mobility analysis.

CERTIFICATIONS

- AWS AI & ML Scholar and AWS AI Practitioner Challenge (Udacity, 2026)
- NPTEL Elite: Introduction to IoT, IIT Kharagpur (75%)